

Inside the Filing Cabinet: How LCSH Authority Records Work and How They Become a Map

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“A subject heading is not just a label. Behind every authorized heading sits a whole dossier—connections to neighboring concepts, notes on intended meaning, and a unique address on the internet. Together, those dossiers form a network that can be turned into a map.”

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What Is an Authority Record?

When a librarian attaches the subject heading “*Banks and banking*” to a book, they are not just typing a label. They are linking that book to an official document called an **authority record**—a structured entry maintained by the Library of Congress that tells the whole story of what that heading means, what it is related to, and when it should be used instead of something else.

Think of an authority record as a profile page for a concept. Just as a social media profile might list your name, a short biography, your connections to other people, and your alternative usernames, an LCSH authority record lists a heading’s official name, notes on its meaning, its connections to other headings, and any informal terms that should redirect to it.

Why this matters for the research: The study’s key measurements— hierarchical distance and category boundaries—are only possible because of the rich relational structure stored in authority records. A system that only stored the heading names would be useless for this kind of analysis. It is the *connections* between headings that make the analysis work.

The Library of Congress publishes all of its authority records freely online at id.loc.gov, where each heading has its own permanent web address (called a URI—Uniform Resource Identifier). For example, “*Banks and banking*” lives at id.loc.gov/authorities/subjects/sh85011534. This persistent address is how the research software fetches each heading’s data automatically.

The Seven Components of an Authority Record

Every fully authorized LCSH heading has up to seven types of information, each identified by a short two-letter code. The figure below shows a real-looking example for the heading “*Cognitive psychology*”, followed by a plain-language explanation of each component.

AUTHORIZED HEADING: Cognitive psychology <i>id.loc.gov/authorities/subjects/sh85026984</i>	
SN	Here are entered works on the branch of psychology that studies mental processes such as perception, memory, and reasoning.
LCC	BF311 (Library of Congress Classification shelf mark)
UF	Mental processes (if someone searches this, they are redirected here)
UF	Cognition—Psychological aspects
BT	Psychology (the broader category this heading belongs to)
NT	Attention (more specific topics that live inside this one)
NT	Memory; Perception; Problem solving (... and more)
RT	Neuropsychology (related, but neither broader nor narrower)
<i>Source: id.loc.gov — illustrative adaptation</i>	

Figure 1: A simplified LCSH authority record for “*Cognitive psychology*.” Each color-coded row corresponds to one type of information. Real records may contain many more NT and UF entries.

Breaking Down Each Tag

Here is what each two-letter code means in plain English, and why it matters for the research:

Tag	Full name	Plain-language meaning and research role
SN	Scope Note	A short note written by catalogers explaining exactly how the heading should be used—and what it should <i>not</i> be used for. Helps the researcher map the correct heading to each word sense.
LCC	LC Classification	The shelf address in a physical library (like a ZIP code for books). Less central to this study, but useful for sanity-checking that headings cover the right subject domain.
UF	Used For	Lists unofficial or alternative names that should all be funneled to this one official heading. This is LCSH’s vocabulary control: whether you search “mental processes” or “cognitive psychology,” the catalog finds the same authorized heading.
USE	Use (see)	The flip side of UF. A non-preferred heading that carries only this one tag, pointing the user to the right place. These records have <i>no</i> BT/NT structure—a critical distinction for the research (see Section 4).
BT	Broader Term	The parent heading—the more general concept this heading lives <i>inside</i> . This is the core ingredient of the research’s hierarchical distance calculation: BT links are the rungs of the knowledge ladder.
NT	Narrower Term	The children headings—more specific topics that live <i>inside</i> this heading. NT is just the downward direction of the same parent–child relationship.
RT	Related Term	A nearby heading that is neither a parent nor a child—more like a cousin. These relationships are noted but are <i>not</i> used in the hierarchical distance calculation, because they do not define a strict up-down tree path.

A Key Asymmetry: BT and NT always come in pairs—if “*Cognitive psychology*” lists “*Psychology*” as a BT, then “*Psychology*” must list “*Cognitive psychology*” as an NT. The relationship is always recorded from both ends. RT relationships also come in pairs. USE and UF also mirror each other. This mutual linking is what makes LCSH a true *network* of concepts, not just an alphabetical list.

Turning Authority Records Into a Map: The LCSH Graph

Once the authority records are collected, the research converts them into a mathematical structure called a **directed graph**—essentially a map where every subject heading is a location, and every BT/NT relationship is a one-way road connecting two locations.

What Is a Directed Graph?

A graph, in mathematics, is simply a collection of **nodes** (dots) connected by **edges** (lines). A *directed* graph means the connections have direction—like a one-way street.

In the LCSH graph:

- Each **node** is one LCSH authority heading.
- Each **directed edge** represents a BT relationship: an arrow pointing *from* a heading *to* its broader parent.
- A heading can have *multiple* broader parents (this is called a polyhierarchy)—unlike a family tree, where you only have two parents.

Why directed? The direction matters because broader-to-narrower is not symmetric. “*Cognitive psychology*” is a narrower term within “*Psychology*,” but “*Psychology*” is not a narrower term within “*Cognitive psychology*.” Using directed arrows lets the software follow the path in the correct upward direction when calculating distances.

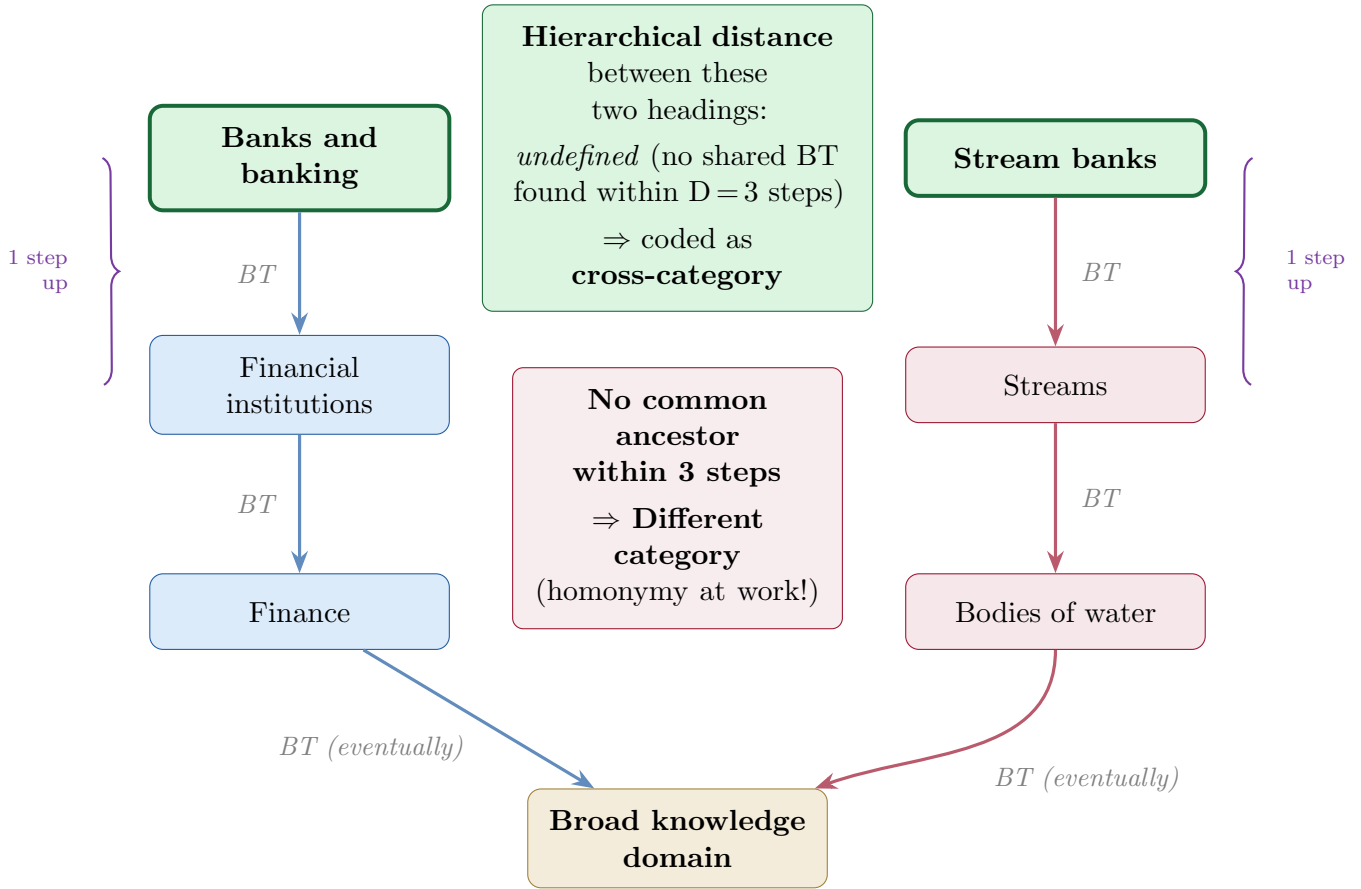


Figure 2: The LCSH graph for the two senses of “bank.” The financial sense (left, blue) and the geographic sense (right, red) live in completely separate branches of the graph with no common ancestor within three steps. This is the clearest possible example of a homonymous word pair: LCSH mirrors the brain’s intuition that these two meanings have nothing to do with each other.

Contrast: A Polysemous Pair That Shares an Ancestor

Unlike homonyms, polysemous words often have senses that are genuinely related. The LCSH graph captures this by showing that their headings share a common ancestor within just a few steps.

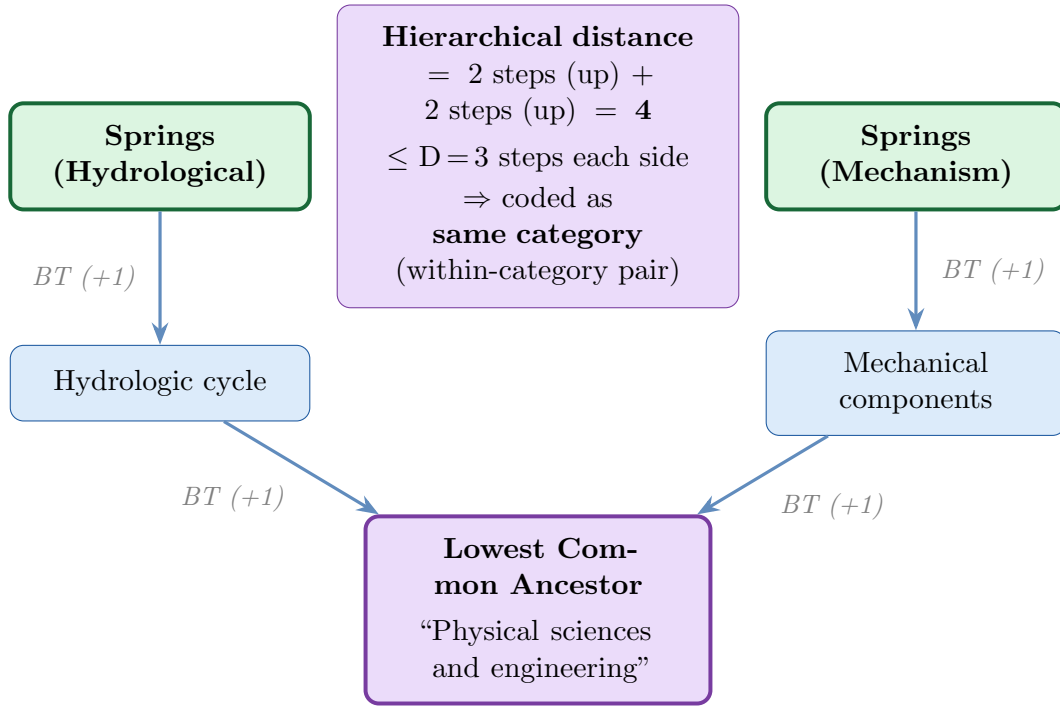


Figure 3: Two senses of the word “spring” (a water spring vs. a mechanical spring) share a common ancestor within three hierarchical steps. The total path length through the lowest common ancestor is 4—this is the *hierarchical distance* value that goes into the statistical models for Research Questions 3 and 4.

Measuring Distance: The Lowest Common Ancestor

To calculate how far apart two headings are in the LCSH graph, the research uses a concept called the **Lowest Common Ancestor (LCA)**—the most specific concept that both headings share as a “grandparent” (or great-grandparent, etc.) in the hierarchy.

An everyday analogy: Imagine two people looking up their family trees. The lowest common ancestor is the most recent common relative. If two cousins share a grandparent, their generational distance through that grandparent is $2 + 2 = 4$ steps. If two people only share a great-great-grandparent, the distance is much larger. In LCSH, every “generation” is one BT step up the hierarchy.

The total hierarchical distance is:

$$d_{\text{hierarchical}} = \underbrace{(\text{steps from Heading 1 to LCA})}_{\text{path up the left branch}} + \underbrace{(\text{steps from Heading 2 to LCA})}_{\text{path up the right branch}}$$

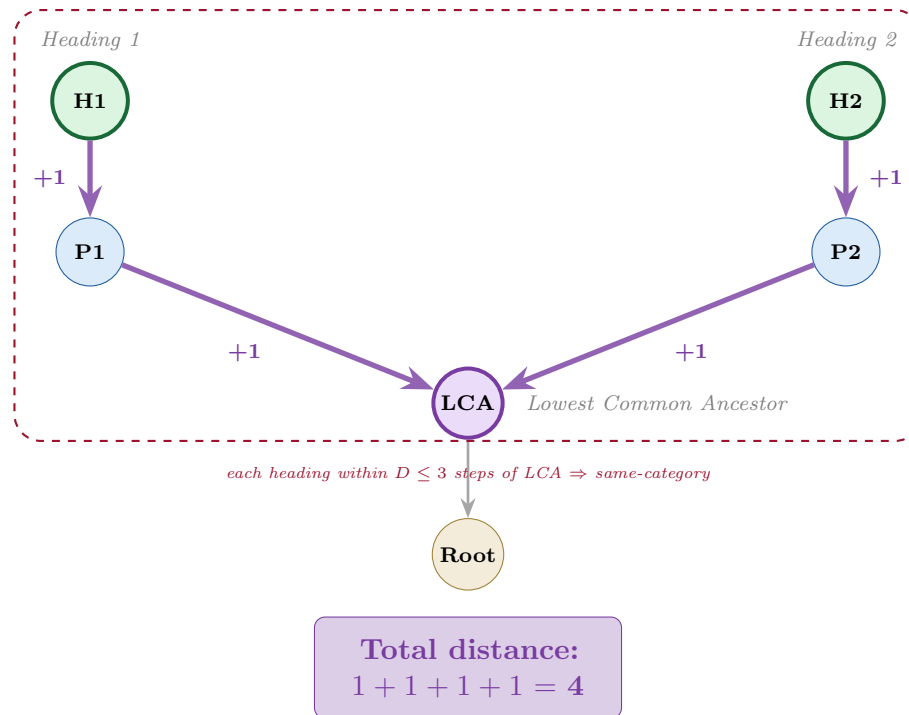


Figure 4: How hierarchical distance is calculated. The software walks up from each heading (green) to the lowest common ancestor (purple), counting steps. The total of all steps is the hierarchical distance. The red dashed box illustrates the $D \leq 3$ threshold: if each heading reaches the LCA in 3 or fewer steps, the pair is classified as same-category (here: 2 steps each, total = 4).

The “D = 3” Rule: Drawing the Neighborhood Boundary

The research needs to decide: how far apart can two headings be in the LCSH hierarchy while still being considered part of the same *conceptual neighborhood*? This threshold is called **D**, and the preregistered default is **D = 3**.

Why 3 steps? Think of it as “second-cousin proximity.” Two LCSH headings that share a common grandparent-level concept are still recognizably in the same subject neighborhood. But headings that only share a great-great-grandparent like “*Knowledge*” or “*Natural sciences*” are too distantly related to count as neighbors—the shared ancestor is so broad it tells us almost nothing.

Three steps also happens to be where most topical LCSH headings live in the hierarchy: typically three to six levels below a disciplinary root like “Psychology” or “Earth sciences.” At depth 3, the common ancestor usually still names a recognizable subject area, not a philosophical abstraction.

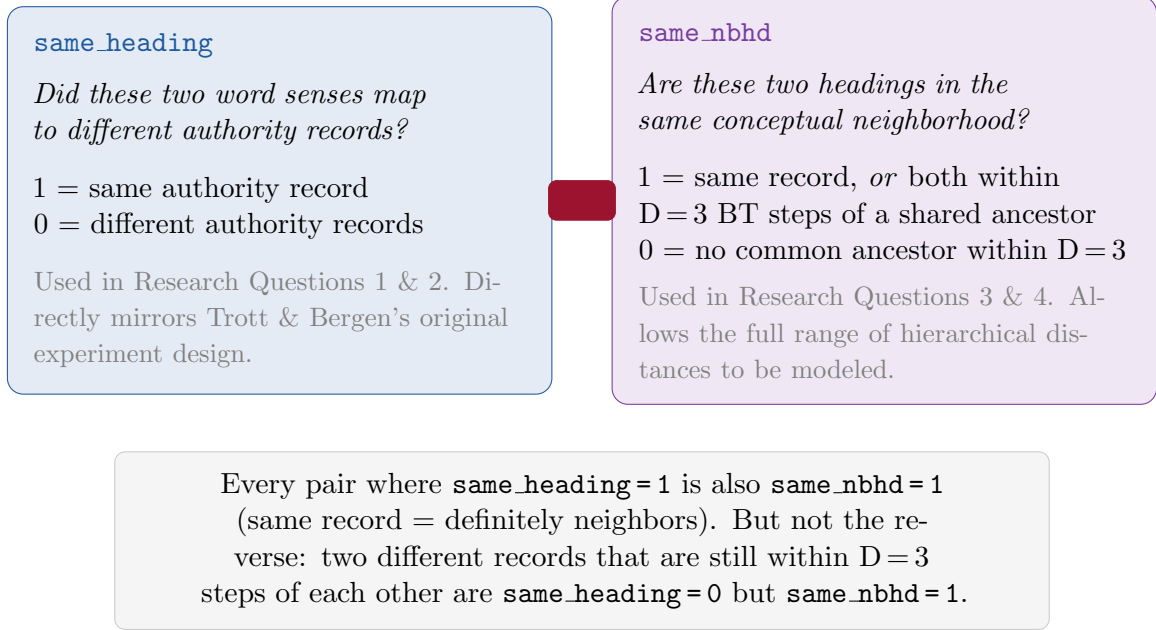


Figure 5: The two categorization variables used in the study. **same_heading** is a simple yes/no: are these the same records? **same_nbhd** is a richer yes/no: are they in the same conceptual neighborhood, accounting for LCSH’s hierarchical structure?

Putting It All Together: From Records to Research

Here is the full pipeline from an authority record to a number in a statistical model, shown as a step-by-step workflow:

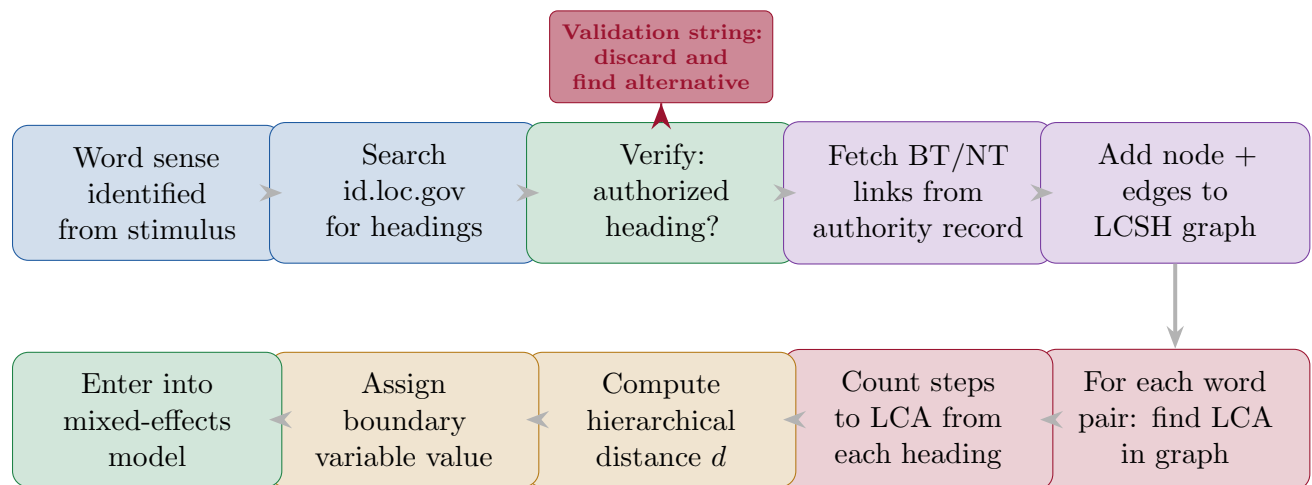


Figure 6: The full pipeline from word sense to statistical predictor. Each word sense goes through manual verification before entering the graph. The graph then produces a numerical distance value for every pair of headings, which feeds directly into the statistical models.

The payoff: After running all 50 word senses ($25 \text{ words} \times 2 \text{ senses each}$) through this pipeline, the result is a dataset where every pair of sentences in the experiment has two additional numbers attached:

- A **BERT distance** (how far apart the two meanings are in AI-measured semantic space)
- A **hierarchical distance** (how far apart the two corresponding LCSH headings are in the graph)

The statistical models then ask: after accounting for the BERT distance, does the hierarchical distance *still* predict how fast participants respond and how highly they rate the LCSH headings? If it does, LCSH's carefully constructed knowledge structure carries genuine cognitive value—above and beyond what any AI language model can capture on its own.